• Medium : #2 : Add Two Numbers

▲ **Problem Statement** : Given two linked lists representing numbers in reverse order, add them & return the sum as a linked list.

▲ **Main Solution** : Add corresponding nodes like normal addition & keep track of the "carry".

▲ **Brute Force** : Convert both linked lists into numbers, add them, then convert the result back into a linked list.

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Algorithm: 1|. Create a dummy node & "carry = 0"

2|. While "L1", "L2", or "carry" exists :-

- Get values from both lists, or "0" if a list ends.
- Calculate "sum = value 1 + value 2 + carry."
- Create a new node with "sum % 10"
- Update "carry = sum / 10"
- Move to the next nodes.

3|. Return "dummy.next".

Important Points:
- Digits are already in reverse order
- Use "carry" for sums greater than "9".
- If one list ends, use "0" for its value.
- "dummy" helps keep track of the answer's starting node.
- Return "dummy.next", not "dummy"

Complexity: Brute Force: $O(n)$ & $O(n)$
Main Solution: $O(n)$ & $O(n)$